General Dessert #4: Phone Operating Procedures

USB vs LSB conventions across HF, what SSB actually is, breaking in, CQ DX etiquette, VOX, and ALC adjustment.

Group G2A · 12 pool questions · Self-study reference · Practice exams at eham.net/exams

Key Facts Reference

USB vs LSB Conventions

Where you are	Sideband	Q
14 MHz and higher (20 m and up)	**Upper sideband (USB).**	G2A01
160 m, 75 m, 40 m bands	**Lower sideband (LSB).**	G2A02
VHF and UHF SSB	**Upper sideband (USB).**	G2A03
17 m and 12 m bands	**Upper sideband (USB).**	G2A04
Why LSB on 160/75/40 m	**Commonly accepted amateur practice** (it's just tradition).	G2A09

What SSB Is and Why We Use It

Question	Answer	Q
Most common HF voice mode	**Single sideband (SSB).**	G2A05
Advantage of SSB over other analog voice modes on HF	**Less bandwidth used and greater power efficiency.**	G2A06
What SSB actually transmits	**Only one sideband; the other sideband and the carrier are suppressed.**	G2A07

Operating Practice

Question	Answer	Q
Recommended way to break into a phone contact	**Say your call sign once.**	G2A08
VOX vs PTT	**VOX allows "hands free" operation.**	G2A10
Who should answer "CQ DX" from a station in the lower 48	**Stations outside the lower 48 states.**	G2A11
Control used to set proper ALC on an SSB rig	**Transmit audio (microphone) gain.**	G2A12

Dessert 4 Question Bank (G2A)

G2A01. Which mode is most commonly used for voice communications on frequencies of 14 MHz or higher?
A. Upper sideband
B. Lower sideband
C. Suppressed sideband
D. Double sideband

G2A02. Which mode is most commonly used for voice communications on the 160-, 75-, and 40-meter bands?
A. Upper sideband
B. Lower sideband
C. Suppressed sideband
D. Double sideband

G2A03. Which mode is most commonly used for SSB voice communications in the VHF and UHF bands?
A. Upper sideband
B. Lower sideband
C. Suppressed sideband
D. Double sideband

G2A04. Which mode is most commonly used for voice communications on the 17- and 12-meter bands?
A. Upper sideband
B. Lower sideband
C. Suppressed sideband
D. Double sideband

G2A05. Which mode of voice communication is most commonly used on the HF amateur bands?
A. Frequency modulation
B. Double sideband
C. Single sideband
D. Single phase modulation

G2A06. Which of the following is an advantage of using single sideband, as compared to other analog voice modes on the HF amateur bands?
A. Very high-fidelity voice modulation
B. Less subject to interference from atmospheric static crashes
C. Ease of tuning on receive and immunity to impulse noise
D. Less bandwidth used and greater power efficiency

G2A07. Which of the following statements is true of single sideband (SSB)?
A. Only one sideband and the carrier are transmitted; the other sideband is suppressed
B. Only one sideband is transmitted; the other sideband and carrier are suppressed
C. SSB is the only voice mode authorized on the 20-, 15-, and 10-meter amateur bands
D. SSB is the only voice mode authorized on the 160-, 75-, and 40-meter amateur bands

G2A08. What is the recommended way to break into a phone contact?
A. Say “QRZ” several times, followed by your call sign
B. Say your call sign once
C. Say “Breaker Breaker”
D. Say “CQ” followed by the call sign of either station

G2A09. Why do most amateur stations use lower sideband on the 160-, 75-, and 40-meter bands?
A. Lower sideband is more efficient than upper sideband at these frequencies
B. Lower sideband is the only sideband legal on these frequency bands
C. Because it is fully compatible with an AM detector
D. It is commonly accepted amateur practice

G2A10. Which of the following statements is true of VOX operation versus PTT operation?
A. The received signal is more natural sounding
B. It allows “hands free” operation
C. It occupies less bandwidth
D. It provides more power output

G2A11. Generally, who should respond to a station in the contiguous 48 states calling “CQ DX”?
A. Any caller is welcome to respond
B. Only stations in Germany
C. Any stations outside the lower 48 states
D. Only contest stations

G2A12. What control is typically adjusted for proper ALC setting on a single sideband transceiver?
A. RF clipping level
B. Transmit audio or microphone gain
C. Antenna inductance or capacitance
D. Attenuator level

Answer Key

G2A01: A	G2A02: B	G2A03: A	G2A04: A
G2A05: C	G2A06: D	G2A07: B	G2A08: B
G2A09: D	G2A10: B	G2A11: C	G2A12: B